THE CATEGORY OF SHEAVES OF THE COARSER TOPOLOGY IS A WEAK SERRE SUBCATEGORY AND R EPSILON_* IS AN EQUIVALENCE ON BOUNDED BELOW COMPLEXES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the weak Serre axioms for A_X are verified by transporting kernels, cokernels and extensions along epsilon_{X,*} and epsilon_X^{-1}, and the equivalence is obtained from the equivalence criterion for bounded below derived categories once the conditions (V_n) for all n are read off from the V-C lemma.

Editorial note. The label lemma-cohomological-descent-general names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement refers to the standing situation of the comparison of two topologies, which is named in the statement and kept as an external pointer with its canonical Stacks tag.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Situation of two topologies on one category and the morphism epsilon. Same chapter, the situation comparing a topology tau' with a coarser topology tau on C and the morphism of sites epsilon : C_tau -> C_tau'. Supplies the categories of sheaves, the functors epsilon_*, epsilon^{-1} and the subcategories A_X, A'_X appearing in the statement.
- Characterisation of weak Serre subcategories. Homology, the lemma listing the exactness conditions characterising a weak Serre subcategory. Supplies the four conditions that are verified for A_X at the start of the proof.
- Equivalence of bounded below derived categories from an equivalence on the hearts. Same chapter, the lemma deducing an equivalence D^+_{A'} -> D^+_A from an equivalence of A' with A and the vanishing of the derived functors on A'. Supplies the final step of the proof, after the conditions (V_n) for all n have been obtained.

Source reference. Original lemma 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Prerequisite pointer invoked at proof line(s) 7377; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0EZF: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7292–7292. Prerequisite pointer invoked at proof

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line(s) 7406; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D7U: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6391–6391. Prerequisite pointer invoked at proof line(s) 7421; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6841–6841. Prerequisite pointer invoked at proof line(s) 7362; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0EZ3. For any X in $\mathcal{C}$ the category $\mathcal{A}_X \subset Ab(\mathcal{C}_\tau/X)$ is a weak Serre subcategory and the functor*

$$R\epsilon_{X,*} : D^+_{\mathcal{A}_X}(\mathcal{C}_\tau/X) \longrightarrow D^+_{\mathcal{A}'_X}(\mathcal{C}_{\tau'}/X)$$

is an equivalence with quasi-inverse given by ϵ_X^{-1}.

Proof. We need to check the conditions listed in Homology, Lemma 0754 for $\mathcal{A}_X$. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a map in $\mathcal{A}_X$, then $\epsilon_{X,*}\varphi : \epsilon_{X,*}\mathcal{F} \to \epsilon_{X,*}\mathcal{G}$ is a map in $\mathcal{A}'_X$. Hence $\mathrm{Ker}(\epsilon_{X,*}\varphi)$ and $\mathrm{Coker}(\epsilon_{X,*}\varphi)$ are objects of $\mathcal{A}'_X$ as this is a weak Serre subcategory of $Ab(\mathcal{C}_{\tau'}/X)$. Applying ϵ_X^{-1} we obtain an exact sequence

$$0 \to \epsilon_X^{-1}\mathrm{Ker}(\epsilon_{X,*}\varphi) \to \mathcal{F} \to \mathcal{G} \to \epsilon_X^{-1}\mathrm{Coker}(\epsilon_{X,*}\varphi) \to 0$$

and we see that $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ are in $\mathcal{A}_X$. Finally, suppose that

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$$

is a short exact sequence in $Ab(\mathcal{C}_\tau/X)$ with $\mathcal{F}_1$ and $\mathcal{F}_3$ in $\mathcal{A}_X$. Then applying $\epsilon_{X,*}$ we obtain an exact sequence

$$0 \to \epsilon_{X,*}\mathcal{F}_1 \to \epsilon_{X,*}\mathcal{F}_2 \to \epsilon_{X,*}\mathcal{F}_3 \to R^1\epsilon_{X,*}\mathcal{F}_1 = 0$$

Vanishing by Lemma 0EZF. Hence $\epsilon_{X,*}\mathcal{F}_2$ is in $\mathcal{A}'_X$ as this is a weak Serre subcategory of $Ab(\mathcal{C}_{\tau'}/X)$. Pulling back by ϵ_X we conclude that $\mathcal{F}_2$ is in $\mathcal{A}_X$.

Thus $\mathcal{A}_X$ is a weak Serre subcategory of $Ab(\mathcal{C}_\tau/X)$ and it makes sense to consider the category $D^+_{\mathcal{A}_X}(\mathcal{C}_\tau/X)$. Observe that $\epsilon_X^{-1} : \mathcal{A}'_X \to \mathcal{A}_X$ is an equivalence and that $\mathcal{F}' \to R\epsilon_{X,*}\epsilon_X^{-1}\mathcal{F}'$ is an isomorphism for $\mathcal{F}'$ in $\mathcal{A}'_X$ since we have (V_n) for all n by Lemma 0EZF. Thus we conclude by Lemma 0D7U. □